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Ontology-Driven Decision Support for Machine Learning Method Selection in Product Development and Production

Combining Semantic Literature Retrieval with Ontology-Based Filtering and Implementation Support

Master's thesis in Engineering and ICT
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June 2026

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ABSTRACT

Write an abstract/summary of your thesis, and state your main findings here.

A summary should be included in both English and any second language, if this is applicable, regardless if the thesis is written in English or in your preferred language. These should be on separate pages, the English version first.

PREFACE

Write the preface of your thesis here.

You may include acknowledgements and thanks as part of your preface on this page, or you may add it as a new chapter after the preface.

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ABBREVIATIONS

List of all abbreviations in alphabetic order:

- **NTNU** Norwegian University of Science and Technology

CHAPTER

ONE

INTRODUCTION

This chapter introduces the purpose and overall direction of the thesis. It presents the motivation for the study and describes the main problem that is addressed. The research questions are then defined to guide the work. Finally, the chapter provides an overview of the structure of the thesis.

1.1 Motivation

Machine learning has seen strong growth in recent years and is now widely applied in product development and production. It is used for tasks such as optimization, control, prediction, and decision support [1].

Advances in data availability and computing power have further increased the use of machine learning in industrial settings [2]. As a result, machine learning is now an important part of modern engineering and manufacturing systems.

At the same time, the field of machine learning is large and diverse. Many different algorithms, methods, and approaches are available, each with its own strengths and weaknesses [1].

There are no clear rules for which method is best in general, and performance depends on the specific application and data [3]. This creates a challenge for engineers and practitioners, who must select suitable methods for their problems.

The large number of available options can act as a barrier and limit the use of machine learning in practice [3].

This challenge is further increased by the gap between research and practical implementation. Research results are spread across many articles, journals, and domains, which makes it difficult to gain an overview [3].

At the same time, companies often lack the knowledge and experience needed to apply machine learning methods effectively [4]. As a result, method selection is often based on trial and error, which can be time-consuming and inefficient [4].

1.2 Problem Description

Selecting appropriate machine learning methods for real-world problems remains a difficult task. Engineers and practitioners often struggle to identify which methods are relevant for their specific problem.

Existing approaches are either too general or too focused on specific techniques, and they do not provide clear guidance for method selection.

A key challenge is the lack of tools that connect problem descriptions, machine learning methods, and supporting research. While many studies focus on developing new models, fewer address how to select methods based on problem characteristics [5].

There is also no standard process for developing machine learning solutions from idea to implementation [6]. This makes it difficult for non-experts to get started.

The large amount of available literature adds to this challenge. Relevant knowledge is distributed across many sources, and it is difficult to identify which articles are useful for a given problem [3].

In addition, there is often no clear connection between the type of data, the problem context, and the selected machine learning method [7].

This can lead to poor choices, such as using complex models on limited data.

There is therefore a need for tools that can:

- support the selection of machine learning methods based on problem descriptions
- provide interpretable recommendations
- link methods to relevant research literature
- support the first steps of implementation

Such a system can help reduce complexity and make machine learning more accessible for engineers and practitioners.

1.3 Research Questions

Based on the challenges described above, this thesis addresses the following research questions:

- How can ontology-based knowledge improve machine learning method selection?
- How can semantic similarity between problem descriptions and literature support recommendations?
- How can structured knowledge and unstructured text be combined in a single system?
- To what extent can such a system support early-stage machine learning implementation?

1.4 Thesis Structure

The remainder of this thesis is structured as follows:

- Chapter 1 introduces the motivation, problem description, and research questions
- Chapter 2 presents background and related work
- Chapter 3 describes the methodology
- Chapter 4 presents the system implementation
- Chapter 5 presents the results
- Chapter 6 discusses the findings and limitations
- Chapter 7 concludes the thesis and suggests future work

CHAPTER

TWO

BACKGROUND AND STATE OF THE ART

2.1 Machine Learning Method Selection

- Overview of ML paradigms (supervised, unsupervised, etc.)
- Typical ML tasks (classification, regression, time series, etc.)
- Challenges in selecting appropriate methods
- Existing guidelines or frameworks for method selection

2.2 Decision Support Systems

- Systems that support decision-making, not replace it
- Knowledge-based vs data-driven systems
- Often interactive and require user input
- Relevance for engineering and industrial contexts

2.3 Implementation Support for Machine Learning

- Selecting a method is only the first step in ML projects
- Users often need support to start implementation
- Templates and reusable workflows can reduce effort
- Such support can lower the barrier for non-experts

2.4 Ontologies and Knowledge Representation

- What an ontology is (concepts, relations, structure)
- Use of ontologies in engineering and ML domains
- Benefits: structure, explainability, reasoning
- Limitations: manual effort, rigidity

2.5 Semantic Retrieval and Text Similarity

- Text can be represented as embeddings (vectors)
- Similarity measures can compare meaning between texts
- Used to match problem descriptions with relevant content
- More flexible than keyword-based search

2.6 Literature-Based Discovery / Retrieval

- Scientific articles are a rich but unstructured knowledge source
- Challenges include large volume, noise, and unclear relevance
- Methods can be linked to articles through extraction and metadata
- Semantic retrieval can improve matching between problems and literature

2.7 Human-in-the-Loop and Feedback

- User input helps guide and refine recommendations
- Feedback can improve system performance over time
- Requires large user base and careful design to avoid bias

2.8 Web-Based ML Tools / Interfaces

- User interfaces allow interaction with the system
- Should support clear input and understandable output
- Transparency helps users trust recommendations
- Focus is on usability, not detailed UX design

2.9 ML in Product Development and Production

- Go through findings from the literature review in pre-master project

- ML is used for tasks such as Quality prediction and defect detection, Process optimization, Predictive maintenance, Time series forecasting
- Product development often has limited and uncertain data
- Production systems may have structured and continuous data
- Selecting suitable ML methods is challenging in these contexts

CHAPTER

THREE

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CHAPTER FOUR

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FIVE

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SEVEN

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REFERENCES

[1] Thorsten Wuest et al. “Machine learning in manufacturing: advantages, challenges, and applications”. In: *Production Manufacturing Research* 4.1 (2016). doi: 10.1080/21693277.2016.1192517, pp. 23–45. ISSN: null. DOI: 10.1080/21693277.2016.1192517. URL: https://doi.org/10.1080/21693277.2016.1192517.

[2] Rahul Rai et al. “Machine learning in manufacturing and industry 4.0 applications”. In: *International Journal of Production Research* 59.16 (2021). doi: 10.1080/00207543.2021.1956675, pp. 4773–4778. ISSN: 0020-7543. DOI: 10.1080/00207543.2021.1956675. URL: https://doi.org/10.1080/00207543.2021.1956675.

[3] Isaac Kofi Nti et al. “Applications of artificial intelligence in engineering and manufacturing: a systematic review”. In: *Journal of Intelligent Manufacturing* 33.6 (2022), pp. 1581–1601. ISSN: 1572-8145. DOI: 10.1007/s10845-021-01771-6. URL: https://doi.org/10.1007/s10845-021-01771-6.

[4] Sebastian Sonntag et al. “A PATTERN LANGUAGE APPROACH TO IDENTIFY APPROPRIATE MACHINE LEARNING ALGORITHMS IN THE CONTEXT OF PRODUCT DEVELOPMENT”. In: *Proceedings of the Design Society* 3 (2023), pp. 365–374. DOI: 10.1017/pds.2023.37. URL: https://www.cambridge.org/core/product/44BBE29AA01344BC5F50174953C5BA8B.

[5] Simone Arena et al. “A conceptual framework for machine learning algorithm selection for predictive maintenance”. In: *Engineering Applications of Artificial Intelligence* 133 (2024), p. 108340. ISSN: 0952-1976. DOI: https://doi.org/10.1016/j.engappai.2024.108340. URL: https://www.sciencedirect.com/science/article/pii/S0952197624004986.

[6] Tingting Chen et al. *Machine Learning in Manufacturing towards Industry 4.0: From ‘For Now’ to ‘Four-Know’*. Electronic Article. 2023. DOI: 10.3390/app13031903.

[7] Dorina Weichert et al. “A review of machine learning for the optimization of production processes”. In: *The International Journal of Advanced Manufacturing Technology* 104.5 (2019), pp. 1889–1902. ISSN: 1433-3015. DOI: 10.1007/s00170-019-03988-5. URL: https://doi.org/10.1007/s00170-019-03988-5.

APPENDICES